

Objective

The objective of Day 7 is to develop a **Simple Password Validator** using Python and Streamlit. The application checks whether a password satisfies basic security requirements such as minimum length, uppercase and lowercase characters, a number, and a special character. This task helps strengthen the understanding of **strings, conditions, loops, character checking, and basic input validation** while creating an interactive web application.

Key Features

- Secure password input using a hidden password field.
- Checks whether the password contains at least **8 characters**.
- Checks for at least one **uppercase letter**.
- Checks for at least one **lowercase letter**.
- Checks for at least one **number**.
- Checks for at least one **special character**.
- Displays individual validation results.
- Clearly identifies requirements that are missing.
- Displays how many validation requirements are satisfied.
- Shows a success message when all requirements are fulfilled.
- Provides a clean and interactive Streamlit interface.

Technologies Used

The project is developed using **Python** and **Streamlit**. Python string methods such as `isupper()`, `islower()`, and `isdigit()` are used to check individual characters. The `string` module is used to identify special characters, while conditional statements and loops are used to perform the validation. Streamlit provides the interactive user interface and allows the password to be entered using a hidden password field.

Working Principle

The application accepts a password from the user through a secure password input field. After the user clicks the validation button, the program checks the password against five predefined requirements. It verifies the length, uppercase character, lowercase character, digit, and special character. Each requirement is evaluated independently, and the

application displays a clear success or error message for each condition. If all five requirements are satisfied, the password is marked as valid; otherwise, the application identifies the requirements that need improvement.

Learning Outcomes

This project provides practical experience in working with **Python strings, character validation, conditional statements, loops, the string module, and Streamlit components**. It also demonstrates how basic programming concepts can be used to create a practical input-validation application.

```
import streamlit as st
import string

st.set_page_config(
    page_title="Password Validator",
    page_icon="🔒",
    layout="centered"
)

st.markdown("""
<style>
    .title {
        text-align: center;
        font-size: 40px;
        font-weight: bold;
        margin-bottom: 5px;
    }

    .subtitle {
        text-align: center;
        font-size: 18px;
        margin-bottom: 30px;
    }

    .result {
        padding: 20px;
        border-radius: 12px;
        margin-top: 20px;
    }

    .requirement {
        padding: 8px;
        margin: 5px 0;
        border-radius: 8px;
    }
</style>
""", unsafe_allow_html=True)

st.markdown(
    '<div class="title">🔒 Password Validator</div>',
    unsafe_allow_html=True
)
```

```
)

st.markdown(
    '<div class="subtitle">'
    'Check whether your password satisfies the required rules'
    '</div>',
    unsafe_allow_html=True
)

st.divider()

password = st.text_input(
    "Enter your password",
    type="password",
    placeholder="Enter password..."
)

st.subheader("📋 Password Requirements")

st.write("Your password should contain:")

st.write("• At least 8 characters")
st.write("• At least one uppercase letter")
st.write("• At least one lowercase letter")
st.write("• At least one number")
st.write("• At least one special character")

if st.button("🔍 Validate Password", use_container_width=True):

    # Conditions
    length_valid = len(password) >= 8
    uppercase_valid = any(char.isupper() for char in password)
    lowercase_valid = any(char.islower() for char in password)
    digit_valid = any(char.isdigit() for char in password)

    special_valid = any(
        char in string.punctuation for char in password
    )
```

OUTPUT:<http://localhost:8501>

Validation Result

✓ Minimum length requirement satisfied

✓ Contains an uppercase letter

✓ Contains a lowercase letter

✓ Contains a number

✓ Contains a special character

 Password is VALID!

✓ Strong Password

Your password satisfies all the required validation rules.

 For security, the password is entered using a hidden password field and is not displayed on the screen.

 VEDA Internship | Python Programming Track | Day 7

Password Validator

Check whether your password satisfies the required rules

Enter your password

Sham123@



Password Requirements

Your password should contain:

- At least 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one number
- At least one special character

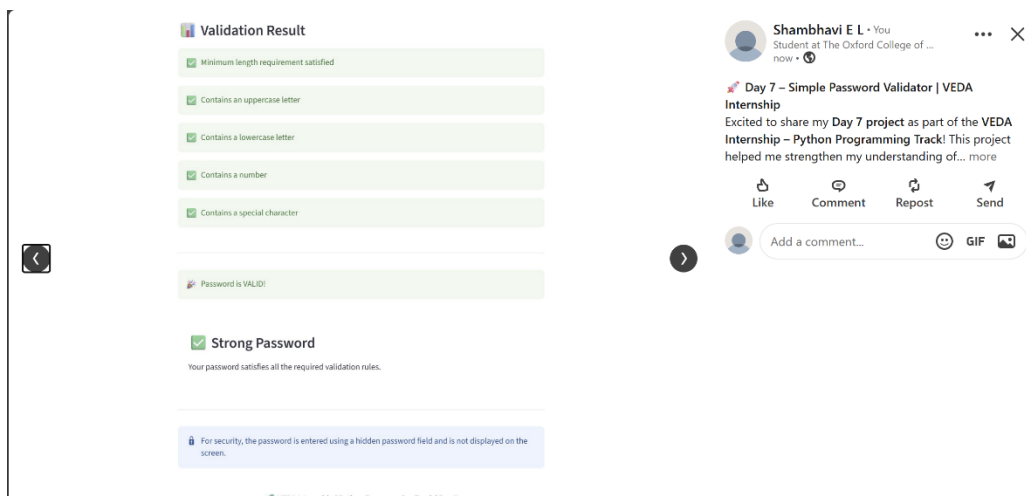
 Validate Password

```
PS C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY7> & 'C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY7\python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '6'

=====
SIMPLE PASSWORD VALIDATOR
=====
Enter your password: shambhavi

-----
PASSWORD VALIDATION RESULT
-----
✓ Minimum length: Passed
X Uppercase letter: Failed
✓ Lowercase letter: Passed
X Number: Failed
X Special character: Failed
-----
X Password is INVALID
Please make sure your password satisfies all requirements.
-----
```

LINKEDIN: <https://lnkd.in/p/dKajpJrt>



Github:<https://github.com/shambhavi0804/VEDAinternship/commit/57a785a638ee5b875a5cef8f084bd3db1d30a07e>

